

Lab: Behavioral Paradigms for Active Inference

Learning Goal: Design, analyze, and evaluate behavioral experiments testing Active Inference predictions.

Part 1: Eye-Tracking Experiment Design

Exercise: Design an eye-tracking study testing epistemic foraging:

Task: Visual scene containing objects at 8 locations. Some locations are familiar (low uncertainty), others novel (high uncertainty). Participant's goal: identify all objects.

Active Inference prediction: First fixations should be directed toward high-uncertainty (novel) locations — epistemic foraging maximizes information gain.

Specify:

1. **Stimulus design:** 8 objects, 4 familiar (seen in training) + 4 novel
2. **Dependent variable:** Proportion of first fixations to novel vs. familiar locations
3. **Expected result:** >70% first fixations to novel locations
4. **Control experiment:** Repeat with explicit reward at specific locations (pragmatic motive) — does pragmatic value override epistemic foraging?
5. **Analysis:** Compute information gain for each fixation sequence and correlate with scanpath data

Write your response here

Part 2: Pupillometry Experiment

Learning Goal: Design a pupillometry study testing precision and surprise.

Exercise: Task = auditory oddball with 3 conditions:

Condition	Standard Probability	Deviant Probability	Expected Pupil Response to Deviant
Predictable	90%	10%	Large dilation (high PE due to strong prediction)
Moderate	70%	30%	Moderate dilation
Unpredictable	50%	50%	Small dilation (weak prediction = small PE)

Prediction: Pupil dilation to deviants scales with prediction strength (more predictable contexts → larger PE → larger pupil dilation).

Additionally measure **tonic pupil size**:

- Unpredictable condition → larger tonic pupils (higher uncertainty / volatility)
- Predictable condition → smaller tonic pupils

Specify the analysis: How do you isolate the phasic (event-related) from tonic pupil components?

Write your response here

Part 3: Reversal Learning Analysis

Learning Goal: Analyze computational signatures in reversal learning data.

Exercise: A participant completes 200 trials of probabilistic reversal learning (2 options, reward reversals every ~30 trials):

Behavioral signatures to identify:

Signature	Description	Computational Interpretation
Win-stay probability	P(repeat choice	previous reward)
Lose-shift probability	P(switch	previous loss)
Post-reversal perseveration	Trials to switch after reversal	Prior volatility ω (low $\omega \rightarrow$ more perseveration)
Pre-reversal switching	Premature switches before reversal	High ω (over-estimating volatility)
Choice consistency	Overall stability of choice patterns	β (policy precision)

Estimate the participant's computational phenotype from these behavioral signatures.

Write your response here

Part 4: Information Sampling Task

Learning Goal: Design a task that measures the exploration-exploitation trade-off.

Exercise: Design a "when to commit" task:

Task: Participant sees 12 boxes. Each box hides a token (red or blue). They can open boxes (information sampling) or commit to guessing which color is majority (decision). Reward if correct.

Active Inference prediction: Agent opens boxes until $EFE(\text{commit}) < EFE(\text{sample})$:

- $G(\text{commit})$ = risk of wrong guess (pragmatic cost)
- $G(\text{sample})$ = cost of time + expected information gain

Manipulation: Vary the cost of sampling (low cost vs. high cost per box opened)

Sampling Cost	Predicted Behavior	AI Parameter
Low (1 point per box)	Open many boxes (~8-10)	High epistemic value threshold
High (5 points per box)	Open few boxes (~3-4)	Lower threshold, accept more risk

How would clinical populations differ? (Jumping-to-conclusions in schizophrenia → open fewer boxes)

Write your response here

Part 5: Reflection

In 300 words, reflect: Behavioral experiments are necessarily indirect — we observe choices, reaction times, and eye movements, not prediction errors or precision. How confident can we be that the computational variables we infer actually correspond to brain processes? What additional evidence (neural, pharmacological) is needed?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Experiment design	Epistemic foraging via eye-tracking
2	Pupillometry design	Precision and surprise measurement
3	Data interpretation	Computational phenotyping from behavior
4	Task design	Exploration-exploitation trade-off
5	Critical reflection	Inference-to-brain mapping